

Unsupervised Brain MRI Anomaly Detection with Enhanced StRegA Integrating Diffusion Models, Multi-Scale Distillation, and Normative Representation Learning

Unsupervised anomaly detection (UAD) in brain MRI is used to identify abnormal aberrations without the necessity for lesion-specific annotations. One famous approach, StRegA, makes the process easier by adding tissue segmentation during preprocessing. This technique allows for the employment of a compact context-encoding variational autoencoder (VAE), decreasing dependency on large architectures while making the most of anatomical context. However, despite the advantages of input simplicity, StRegA has major restrictions. Its reliance on a VAE might result in too smooth reconstructions that overlook small lesions, and flaws in tissue segmentation can obscure or misclassify diseases.

To address these challenges, current techniques have shown promising results. Scale-Aware Contrastive Reverse Distillation (SCRD), for example, improves feature discriminability at many scales, allowing for more accurate detection of anomalies regardless of size. Also, compared to VAEs, both conditional and guided diffusion models produce sharper pseudo-healthy reconstructions that keep normal structure while drawing attention to abnormal areas. Techniques like as masked diffusion pretraining (MAD-AD) allow models to reconstruct missing image patches, improving their capacity to pinpoint anomalies. Furthermore, normative modelling focusses on reflecting the normal variability of healthy anatomy, which helps to reduce false positives. Although these techniques have performed well on benchmark datasets, their efficacy on varied clinical groups is little unknown.

In this work, we suggest improving StRegA by combining its input-simplification technique with these recent advancements. We want to see if giving tissue-aware context may still improve the system's robustness against scanner variability and anatomical variation when combined with diffusion and contrastive models, while also addressing the inherent risks of preprocessing reliance. Our full pipeline is built and validated utilising three datasets, each of which plays a complimentary role: IXI for healthy anatomy modelling, MOOD for controlled testing with synthetic anomalies, and BraTS for real tumour assessments.

Our research has a double impact. On the scientific front, it integrates multiple state-of-the-art methodologies into a unified anomaly detection framework and examines how traditional simplification strategies interact with modern generative and contrastive learning techniques. Clinically, it represents progress towards more reliable neuroimaging tools by improving reconstruction fidelity, feature discriminability, and variability robustness—bringing automated anomaly detection closer to practical clinical deployment, where systems must identify unexpected abnormalities without relying on pathology-specific training data.